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riya.soni17oct@gmail.com ▾

NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Data Science for Engineers (course)

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to check your
payment status

Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 1 ()

Week 2 ()

- ☐ Linear Algebra for Data science (unit? unit=36&lesson=37)
- ☐ Solving Linear Equations (unit? unit=36&lesson=38)
- ☐ Solving Linear Equations (

Week 2 : Assignment 2

The due date for submitting this assignment has passed.

Due on 2025-08-06, 23:59 IST.

As per our records you have not submitted this assignment.

- 1) Are the vectors $\begin{bmatrix} -2 \\ 4 \end{bmatrix}$, $\begin{bmatrix} 7 \\ -2 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ -6 \end{bmatrix}$ linearly independent? **1 point**

- ☐ Yes
☐ No

No, the answer is incorrect.

Score: 0

Accepted Answers:

No

- 2) Does the set, $S = (1, 1), (1, 2)$ spans $\mathbb{R}^2$? **1 point**

- ☐ Yes
☐ No

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes

- 3) Consider the following system of linear equations of the form $Ax = b$: **1 point**

$$2x - 3y + 6z = 14$$

$$x + y - 2z = -3$$

Which among the following are correct?

☐ $\begin{bmatrix} 1 \\ -4 \\ 0 \end{bmatrix}$ is a solution to $Ax = b$

Continued)
(unit?
unit=36&less
n=39)

- ☐ Linear Algebra
- Distance, Hyp
erplanes and
Halfspaces, Eig
envalues, Eig
envectors (unit?
unit=36&less
n=40)

- ☐ Linear Algebra
- Distance, Hyp
erplanes and
Halfspaces, Eig
envalues, Eig
envectors (Continued 1)
(unit?
unit=36&less
n=41)

- ☐ Linear Algebra
- Distance, Hyp
erplanes and
Halfspaces, Eig
envalues, Eig
envectors (Continued 2)
(unit?
unit=36&less
n=42)

- ☐ Linear Algebra
- Distance, Hyp
erplanes and
Halfspaces, Eig
envalues, Eig
envectors (Continued 3)
(unit?
unit=36&less
n=43)

- ☐ Common
doubts asked
on Linear
Algebra (unit?
unit=36&less
n=44)

☐ $\begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$ is a solution to $Ax = b$

☐ $\begin{bmatrix} 1 \\ -4 \\ 0 \end{bmatrix}$ is a solution to $Ax = 0$

☐ $\begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$ is a solution to $Ax = 0$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\begin{bmatrix} 1 \\ -4 \\ 0 \end{bmatrix}$ is a solution to $Ax = b$
 $\begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$ is a solution to $Ax = 0$

Consider the following system of linear equation:

$$\begin{aligned} x + y + z &= -2 \\ x + 2y - z &= 1 \\ 2x + ay + bz &= 2 \end{aligned}$$

4) Find the conditions on a and b for which the above system has no solution.

1 point

- ☐ $2a + b - 6 = 0$
☐ $a \neq 4, 2a + b - 6 = 0$
☐ $a = 4, b = -2$
☐ $2a + b - 6 \neq 0$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$a \neq 4, 2a + b - 6 = 0$

5) Find the conditions on a and b for which the above system has a unique solution.

1 point

- ☐ $2a + b - 6 = 0$
☐ $a \neq 4, 2a + b - 6 = 0$
☐ $a = 4, b = -2$
☐ $2a + b - 6 \neq 0$

No, the answer is incorrect.



Week 2
Feedback
Form : Data
Science for
Engineers!!
(unit?
unit=36&lesso
n=154)

Quiz: Week 2
: Assignment
2
(assessment?
name=237)

Week 3 ()

Week 4 ()

Week 5 ()

Week 6 ()

Week 7 ()

Text
Transcripts
()

Download
Videos ()

Books ()

Problem
Solving
Session -
July 2025 ()

Score: 0

Accepted Answers:

$$2a + b - 6 \neq 0$$

6) Find the conditions on a and b for which the above system has infinite number of solutions.

1 point

☐

$$2a + b - 6 = 0$$

☐

$$a \neq 4, 2a + b - 6 = 0$$

☐

$$a = 4, b = -2$$

☐

$$2a + b - 6 \neq 0$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$a = 4, b = -2$$

In solving the system $Ax = b$ in the variables x_1, x_2, x_3 and Gaussian elimination on the Augmented matrix $[A|b]$ led to the following row echelon form

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 3 & 2 \\ 0 & 1 & 1 & 2 & 3 \\ 0 & 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

7) Identify the number of free variable from the above row echelon matrix.

1 point

☐ 0

☐ 1

☐ 2

☐ 3

No, the answer is incorrect.

Score: 0

Accepted Answers:

1

8) Which among the following is correct for the above system $Ax = b$?

1 point

☐

It has infinite number of solutions.

☐

It has a unique solution.

☐

It has no solution.

No, the answer is incorrect.

Score: 0

Accepted Answers:

It has infinite number of solutions.

9) For what values of a are matrix $A = \begin{bmatrix} a & 1 \\ -2 & a+3 \end{bmatrix}$ not invertible?

1 point

- ☐ $a = 1$
- ☐ $a = -2$
- ☐ $a = -1$
- ☐ $a = 2$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$a = -2$

$a = -1$

10) Which among the following is true for the determinant of a matrix?

1 point

- ☐ The determinant of a diagonal matrix is the product of its diagonal entries.
- ☐ If one row of a matrix is a scalar multiple of another, the determinant is 1.
- ☐ If one row of a matrix is a scalar multiple of another, the determinant is 0.
- ☐ The determinant of a permutation matrix can only be 1.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The determinant of a diagonal matrix is the product of its diagonal entries.

If one row of a matrix is a scalar multiple of another, the determinant is 0.

11) Which among the following are the eigenvalues of matrix

1 point

$$A = \begin{bmatrix} 5 & 8 & 16 \\ 4 & 1 & 8 \\ -4 & -4 & -11 \end{bmatrix}$$

- ☐ 1, 3, -3
- ☐ 1, 3, 3
- ☐ -1, 3, 3
- ☐ 1, -3, -3

No, the answer is incorrect.

Score: 0

Accepted Answers:

1, -3, -3

12) Find the nullity of $A = \begin{bmatrix} 1 & -3 & -2 & 4 \\ 1 & -3 & 1 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

13)

1 point

- -

Let $A = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}$. Suppose the eigen values corresponding to AA^T are a, b and c , then find the value of $ab + bc + ca$.

- ☐ 9
- ☐ 0
- ☐ 81
- ☐ 18

No, the answer is incorrect.

Score: 0

Accepted Answers:

0

